

Pakhi Banchalia

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RESEARCH EXPERIENCE

- **Lossfunk**

July 2025 - Sept 2025

Research Intern

Bangalore, India

- Investigated homeostatic learning in the Crafter environment
- Analyzed agent behavior under varying internal reward structures and compared against baseline approaches

- **Computational Machinery of Cognition (CMC) lab, TU Dresden**

Nov 2024 - June 2025

Research Intern

Dresden, Germany

- Designed homeostatic reinforcement learning agents to study task generalization and retention under internal drive constraints
- Implemented and evaluated Dopamine, a gradient-free learning algorithm based on reward-modulated plasticity

- **Google Summer of Code, INCF**

May 2024 - Sept 2024

Contributor

Remote

- Simulated *C. elegans* neural development using Neural Cellular Automata and evolutionary optimization
- Modeled neural connectivity with graph neural networks
- Applied Kepler Mapper (topological data analysis) to analyze developmental trajectories
- Produced open-source, reproducible research [code](#)

PUBLICATIONS

- Pillai, H.S., **Banchalia, P.**, Jamboji, V., Krishnan, S.R. (2026). *CAN: Continuously Adapting Network*. In *Data Science and Applications, ICDSA 2025*. Lecture Notes in Networks and Systems, vol 1723. Springer, Cham. [\[DOI\]](#)

RESEARCH PROJECTS

- **Homeostatic Crafter**

- Explored Homeostatic Reinforcement Learning in Craftax environment
- Observed agent behaviour in different reward settings
- Tried introducing modularity to further improve agent performance

- **Dopamine: Reward-Modulated Gradient-Free Learning**

- Contributed to a biologically inspired learning rule using reward-modulated weight perturbation
- Explores credit assignment without backpropagation
- Achieved competitive performance in preliminary benchmarks

- **SEANN : Self-Expanding and Adaptive Neural Networks**

- Proposed a neural architecture that dynamically grows capacity for continual learning
- Identifies task-relevant subnetworks to reduce catastrophic forgetting

TECHNICAL SKILLS

- **Languages:** Python, C++, MATLAB, SQL
- **ML / Research:** PyTorch, TorchVision, Gymnasium (OpenAI Gym), JAX (familiar)
- **Scientific Computing:** NumPy, SciPy, Pandas, Matplotlib, Kepler Mapper (TDA)
- **Research Areas:** Reinforcement Learning, Biologically-Inspired Learning, Neural Cellular Automata, Continual Learning, GNNs
- **Developer Tools:** Git, Linux/Unix, Docker, LaTeX, Weights Biases (WB)

EDUCATION

- **Technische Universität Dresden**

Master of Science in Computational Modeling and Simulation (Applied AI track)

- GPA: not yet available

Oct 2025 - Present

Dresden, Germany

- **Amrita Vishwa Vidyapeetham Kerala, India**

Bachelor of Technology in Computer Science and Artificial Intelligence

- Grade: 8.8/10

2021-2025

Amritapuri, India